1. Python Code: Pearson, Spearman, Kendall Correlation

```
import numpy as np

import pandas as pd

from scipy.stats import pearsonr, spearmanr, kendalltau


# -----------------------

# SAMPLE DATA (replace with your own)

# -----------------------

np.random.seed(42)

x = np.random.randint(1, 100, 20)

y = np.random.randint(1, 100, 20)


df = pd.DataFrame({'X': x, 'Y': y})

print(df)


# -----------------------

# 1. PEARSON CORRELATION

# -----------------------

pearson_corr, pearson_p = pearsonr(df['X'], df['Y'])

print("\nPearson Correlation:", round(pearson_corr, 4))

print("Pearson p-value:", round(pearson_p, 4))


# -----------------------

# 2. SPEARMAN CORRELATION

# -----------------------

spearman_corr, spearman_p = spearmanr(df['X'], df['Y'])
```

```
print("\nSpearman Correlation:", round(spearman_corr, 4))
print("Spearman p-value:", round(spearman_p, 4))

# -----------------------
# 3. KENDALL CORRELATION
# -----------------------
kendall_corr, kendall_p = kendalltau(df['X'], df['Y'])
print("\nKendall Correlation:", round(kendall_corr, 4))
print("Kendall p-value:", round(kendall_p, 4))

# -----------------------
# 4. CORRELATION MATRIX (all 3)
# -----------------------
print("\nCorrelation Matrix (Pandas - Pearson by default):")
print(df.corr(method='pearson'))

print("\nCorrelation Matrix - Spearman:")
print(df.corr(method='spearman'))

print("\nCorrelation Matrix - Kendall:")
print(df.corr(method='kendall'))
```

2. Pearson Correlation – Scatter Plot with Best-Fit Line

```
import numpy as np
import pandas as pd
from scipy.stats import pearsonr
```

```
import matplotlib.pyplot as plt

# Sample Data
np.random.seed(10)
x = np.random.randint(1, 100, 30)
y = 2.5 * x + np.random.normal(0, 20, 30)   # linear relationship

# Pearson correlation
r, p = pearsonr(x, y)
print("Pearson Correlation:", r)

# Plot
plt.figure(figsize=(7,5))
plt.scatter(x, y)
plt.title(f"Pearson Correlation (r = {round(r,3)})")
plt.xlabel("X")
plt.ylabel("Y")

# Best fit line
m, b = np.polyfit(x, y, 1)
plt.plot(x, m*x + b)

plt.show()
```

3. Spearman Correlation – Monotonic Relationship Graph

```
import numpy as np
import pandas as pd
```

```
from scipy.stats import spearmanr

import matplotlib.pyplot as plt


# Sample Data (monotonic but non-linear)

np.random.seed(20)

x = np.linspace(1, 50, 30)

y = np.log(x) * 10 + np.random.normal(0, 1, 30)


# Spearman correlation

rho, p = spearmanr(x, y)

print("Spearman Correlation:", rho)


# Plot

plt.figure(figsize=(7,5))

plt.scatter(x, y)

plt.title(f"Spearman Correlation (ρ = {round(rho,3)})")

plt.xlabel("X")

plt.ylabel("Y")

plt.show()
```

4. Kendall Correlation – Concordant vs Discordant Pair Visualization

```
import numpy as np

import pandas as pd

from scipy.stats import kendalltau

import matplotlib.pyplot as plt


# Sample Data
```

```
np.random.seed(30)

x = np.arange(1, 21)

y = np.array([5, 8, 7, 10, 12, 13, 11, 15, 16, 18,

            20, 19, 22, 24, 23, 26, 28, 27, 29, 30])


# Kendall correlation

tau, p = kendalltau(x, y)

print("Kendall Tau:", tau)


# Plot

plt.figure(figsize=(7,5))

plt.scatter(x, y)

plt.title(f"Kendall Correlation (τ = {round(tau,3)})")

plt.xlabel("X")

plt.ylabel("Y")

plt.show()
```

5.pearson correlation heatmap

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns


# Sample dataset

np.random.seed(42)

df = pd.DataFrame({

   "Height": np.random.normal(165, 10, 50),
```

```
    "Weight": np.random.normal(65, 8, 50),
    "Age": np.random.randint(20, 50, 50),
    "Income": np.random.normal(50000, 8000, 50),
})

# Pearson correlation
corr = df.corr(method="pearson")

# Heatmap
plt.figure(figsize=(7, 5))
sns.heatmap(corr, annot=True, cmap="coolwarm", linewidths=0.5)
plt.title("Pearson Correlation Heatmap")
plt.show()
```

6.Spearman Correlation Heatmap

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Sample dataset
np.random.seed(42)
df = pd.DataFrame({
    "X": np.linspace(1, 100, 50),
    "Y": np.log(np.linspace(1, 100, 50)) + np.random.normal(0, 0.1, 50),
    "Z": np.sqrt(np.linspace(1, 100, 50)),
```

```
    "W": np.random.exponential(2, 50)
})

# Spearman correlation
corr = df.corr(method="spearman")

# Heatmap
plt.figure(figsize=(7, 5))
sns.heatmap(corr, annot=True, cmap="viridis", linewidths=0.5)
plt.title("Spearman Correlation Heatmap")
plt.show()
```

7. Kendall Correlation Heatmap

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Sample dataset
np.random.seed(42)
df = pd.DataFrame({
    "Quality": np.random.randint(1, 10, 40),
    "Rating": np.random.randint(1, 10, 40),
    "Rank": np.random.randint(1, 10, 40),
    "Score": np.random.randint(1, 10, 40)
})
```

```
# Kendall correlation
corr = df.corr(method="kendall")


# Heatmap
plt.figure(figsize=(7, 5))
sns.heatmap(corr, annot=True, cmap="magma", linewidths=0.5)
plt.title("Kendall Correlation Heatmap")
plt.show()
```

8.combined heat map

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns


# -----------------------
# Sample Dataset
# -----------------------
np.random.seed(42)
df = pd.DataFrame({
    "A": np.random.normal(50, 10, 60),
    "B": np.random.normal(30, 5, 60),
    "C": np.random.uniform(1, 100, 60),
    "D": np.linspace(10, 100, 60) + np.random.normal(0, 5, 60)
})
```

```
# -----------------------
# Correlations
# -----------------------
corr_pearson = df.corr(method="pearson")
corr_spearman = df.corr(method="spearman")
corr_kendall = df.corr(method="kendall")

# -----------------------
# Plot side-by-side heatmaps
# -----------------------
plt.figure(figsize=(18, 5))

# Pearson
plt.subplot(1, 3, 1)
sns.heatmap(corr_pearson, annot=True, cmap="coolwarm", vmin=-1, vmax=1)
plt.title("Pearson Correlation")

# Spearman
plt.subplot(1, 3, 2)
sns.heatmap(corr_spearman, annot=True, cmap="viridis", vmin=-1, vmax=1)
plt.title("Spearman Correlation")

# Kendall
plt.subplot(1, 3, 3)
sns.heatmap(corr_kendall, annot=True, cmap="magma", vmin=-1, vmax=1)
plt.title("Kendall Correlation")
```

```
plt.tight_layout()

plt.show()
```